

Project Title

Fake QR Scanner (QR Security Risk Analyzer)

Student Name: Rohan Makwana

Enrollment No.: 92301733053

1. Features

1.1 Live Camera Scanner

The application uses **Android Jetpack CameraX** and **Google ML Kit** to scan QR codes in real time through the device camera. When a QR code is detected, its content is automatically extracted and sent for security analysis.

1.2 Security Dashboard

The Security Dashboard retrieves real-time statistics from the **Room Database** and displays important scanning information, including:

- Total Scans
- Safe Scans
- Suspicious Scans
- High-Risk Scans

This provides users with a quick overview of their QR scanning activity and detected security threats.

1.3 Heuristic Risk Analysis

The application uses a heuristic-based security analysis system with **16+ security indicators** to evaluate QR code content and generate a security score.

Examples of indicators include:

- IP-based URLs
- URL shorteners
- Suspicious URL patterns
- Brand impersonation

- Suspicious domains
- Unusual URL structures
- Other potentially risky characteristics

Based on the detected indicators, the application generates a **security score from 0 to 100** and determines an appropriate risk level.

1.4 Gallery Scanning

Users can select an image containing a QR code directly from their phone's gallery. The application detects and extracts the QR code from the selected image and performs the same security analysis.

This feature is useful for analyzing QR codes received as screenshots or saved images.

1.5 Manual Input

Users can manually enter text or a URL into the application. The entered content is then analyzed using the heuristic risk-analysis engine to determine its security risk level.

1.6 Scan History

The application stores previously analyzed QR codes in a **Room Database**. Users can access their previous scans through the Scan History section.

The stored history provides an offline record of previous scans and their analysis results.

1.7 Distribution Chart

The Security Dashboard includes a **risk distribution chart** that provides a visual comparison of different risk levels.

It allows users to easily understand the distribution of:

- Safe scans
- Suspicious scans
- High-risk scans

1.8 Phishing Protection

When a QR code is classified as high risk, the application displays a **strong warning dialog before opening the associated link**.

This provides an additional layer of protection against potentially malicious or phishing websites and allows the user to make an informed decision before proceeding.

2. Technology Stack

Programming Language

Kotlin

Kotlin is used as the primary programming language for developing the Android application using modern Android development practices.

UI Engine

XML with ViewBinding and Material 3

The user interface is developed using XML layouts with ViewBinding. Material 3 components are used to create a modern and consistent user interface with a **Dark Theme**.

ML Engine

Google ML Kit Vision — Barcode Scanning

Google ML Kit is used to detect and decode QR codes and other supported barcodes from camera frames and images.

Camera

Android Jetpack CameraX

CameraX is used to provide camera preview and real-time image analysis. The application uses CameraX's:

- ImageAnalysis
- Preview
- Frame processing
- Backpressure management

capabilities for efficient QR code scanning.

Database

Room Persistence Library (SQLite)

Room is used as the local database solution. It provides persistent offline storage for scan history and allows the application to retrieve stored records for dashboard statistics.

Architecture

MVVM (Model-View-ViewModel) with LiveData

The application follows the MVVM architecture pattern to separate the user interface, application logic, and data management.

LiveData is used to observe database and application data and update the user interface when the underlying data changes.

Development Tools

- **Android Studio**
 - **Gradle**
 - **Android SDK**
-

3. Screenshots and Description

Screen 1: Home Screen / Security Dashboard

The Home Screen provides an overview of the application's security scanning activity.

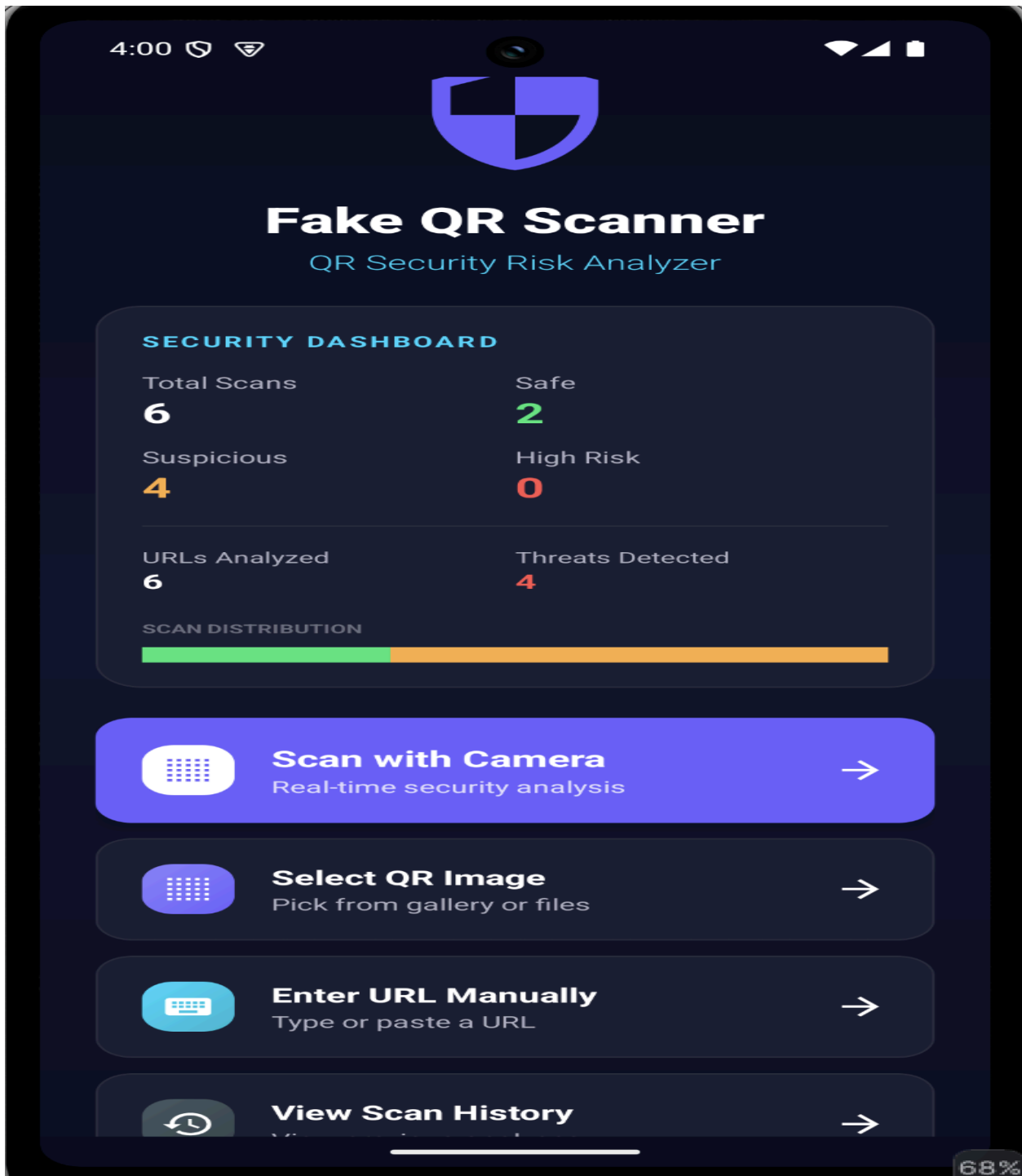
It displays live statistics retrieved from the Room Database. Users can see the total number of scans and the distribution of safe, suspicious, and high-risk scans.

The dashboard also provides a visual comparison of risk levels through the distribution chart.

Description:

The Home Screen displays live security statistics obtained from the Room Database. It provides users with a quick overview of total scans, safe scans, suspicious scans, and detected high-risk threats.

Screenshot:



Screen 2: Live Scanner

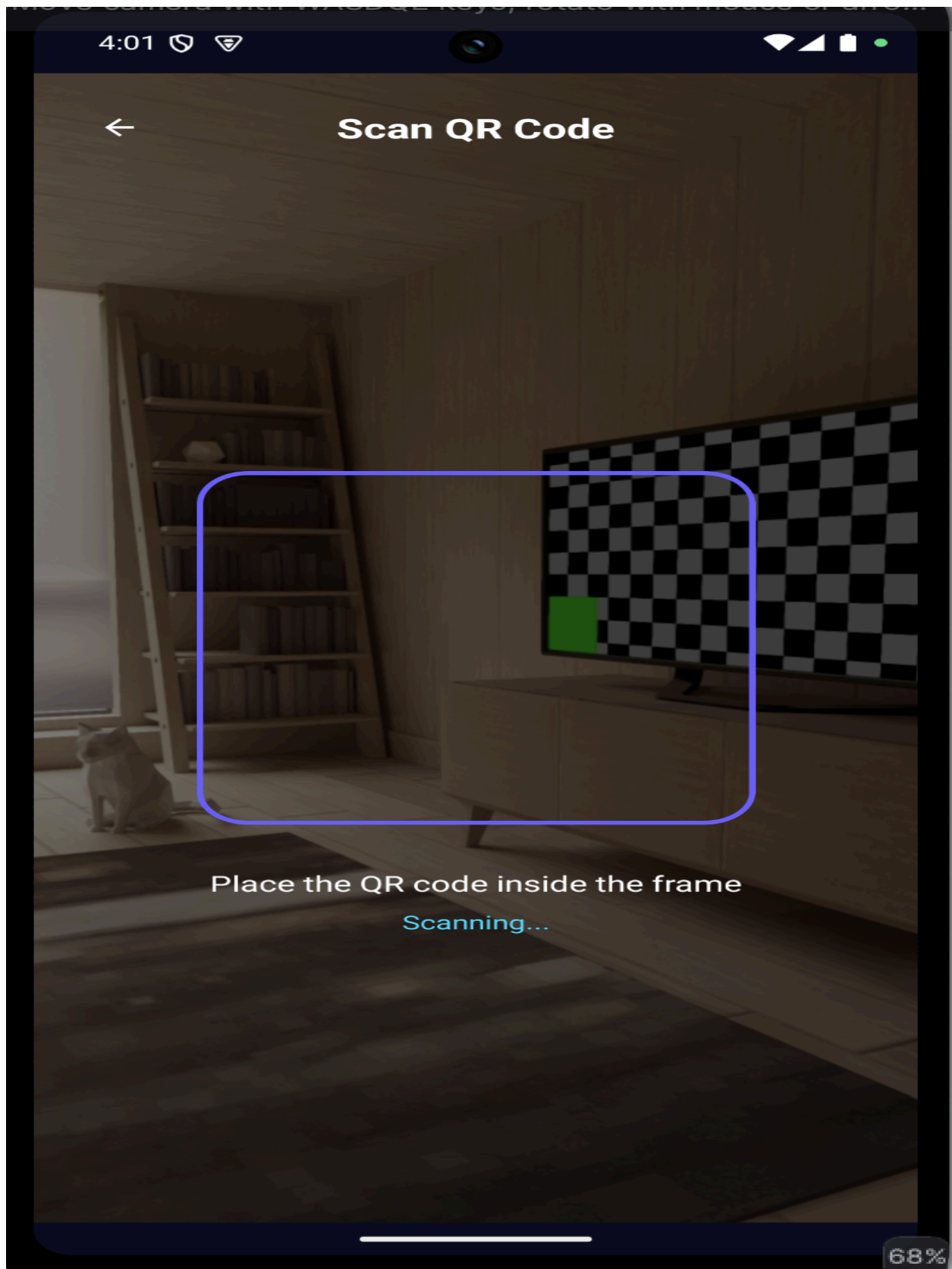
The Live Scanner provides a real-time camera interface for scanning QR codes.

A professional scanning frame is displayed in the center of the camera preview to guide the user. As soon as a QR code is detected, the application automatically extracts its content and sends it to the security analysis engine.

Description:

The Live Scanner uses CameraX for the camera preview and Google ML Kit for real-time QR code detection. Once a QR code is detected, its content is automatically sent for security analysis.

Screenshot:



Screen 3: Analysis Result

The Analysis Result screen displays the complete security analysis of the scanned QR code.

It provides:

- Security Score (0–100)
- Risk Level
- Security Indicators
- Analyzed QR/URL content

The risk level can be classified as **Safe, Suspicious, or High Risk**, depending on the detected security indicators and calculated score.

Description:

The Analysis Result screen displays the security score, risk level, and detected security indicators. It helps users understand why a QR code or URL has been classified as safe, suspicious, or high risk.

16:04



VoLTE 5G

50



Analysis Result

20

/ 100

LOW RISK

Sept 09, 2026 • 04:04 pm

QR CONTENT

<http://example.com/login>

DOMAIN

example.com



HTTPS STATUS

Insecure (HTTP)

SECURITY INDICATORS

- Insecure Connection: Uses HTTP instead of HTTPS — data is transmitted unencrypted
- Suspicious Path Keywords: URL path/query

Screen 4: Scan History

The Scan History screen displays previously analyzed QR codes.

Each record is stored locally in the Room Database and can include information such as the scanned content, risk level, score, and timestamp.

Description:

The Scan History screen provides users with a timestamped record of previously analyzed QR codes. The data is stored locally using Room Database, allowing users to access their scan history offline.

Screenshot:

16:05



Scan History



LOW RISK

Risk Score: 20/100



<http://example.com/login>

example.com

Sept 09, 2026 • 04:04 pm

SUSPICIOUS

Risk Score: 35/100



<https://xn--pple-43d.example/login>

xn--pple-43d.example

Sept 09, 2026 • 04:04 pm

SUSPICIOUS

Risk Score: 50/100



<http://192.0.2.123/admin/login>

192.0.2.123

Sept 09, 2026 • 04:04 pm

4. Future Scope

4.1 API Integration

The application can be enhanced by integrating external threat-intelligence services such as **Google Safe Browsing** and **VirusTotal**.

These APIs can provide real-time URL reputation checks and help identify URLs that are already known to be malicious or associated with phishing activities.

4.2 Deep Link Analysis

Future versions can analyze Android deep links and intent filters to identify potentially dangerous redirections.

This can help prevent malicious QR codes from redirecting users toward unsafe applications or potentially unwanted app installations.

4.3 Dark Web Check

A future version could integrate authorized threat-intelligence or security databases to determine whether a URL, domain, or related indicator has been associated with known data leaks or underground threat activity.

This feature would provide an additional layer of security intelligence.

4.4 Multi-Language Support

The application can be expanded to support multiple regional languages.

Possible languages include:

- English
- Hindi
- Gujarati

Multi-language support would make the application easier to use for a wider range of users.



Certificate

This is to certify Mr./Ms. ROHAN MAHWANA
Studying in Semester 7, Class, SEI Having Roll No. 9230173353
has satisfactorily completed his / her term work in the subject (Name & Code) mPC LOICT0716 for the term
ending in the month of _____, Academic year 20 26 -20 27

Date : 9/9/26

Faculty In-Charge

Name : S m Alam

Signature : Asw

Head of Department : _____





Certificate

This is to certify Mr./Ms. ROHAN MARWADA

Studying in Semester 7, Class, 2KI Having Roll No. 92301733533

has satisfactorily completed his / her term work in the subject (Name & Code) _____

mpc (01CT0716) for the term

ending in the month of _____, Academic year 20 26-2027

Date : _____

Faculty In-Charge

Name : S m Alam

Signature : [Signature]

Head of Department : _____



6. Conclusion

The **Fake QR Scanner (QR Security Risk Analyzer)** is an Android-based security application designed to help users identify potentially risky QR codes before opening their associated content.

The application combines **CameraX, Google ML Kit, Kotlin, Room Database, MVVM, LiveData, and heuristic security analysis** to provide a complete QR security analysis solution.

The major capabilities of the application include real-time QR scanning, gallery scanning, manual URL analysis, heuristic risk scoring, scan history, security statistics, risk distribution visualization, and phishing protection.

The project demonstrates the practical application of **Android development and cybersecurity concepts** to address the growing security risks associated with QR-code-based links.

Future enhancements such as threat-intelligence APIs, deep-link analysis, advanced phishing detection, and multi-language support can further improve the application's security capabilities and usability.